

Impact of vegetation selection on nitrogen and phosphorus processing in bioretention containers

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Received 1 May 2019; Revised 9 July 2019; Accepted 31 July 2019

Maryland State Highway Administration

Additional Supporting Information may be found in the online version of this article.

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Published online 17 September 2019 in Wiley Online Library (wileyonlinelibrary.com)

DOI: 10.1002/wer.1195

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• Abstract

A year-long bioretention container study in Maryland, USA, measured the relationship between three plant species (*Eutrochium dubium*, *Iris versicolor*, and *Juncus effusus*) and N (NO_3^- , NO_2^- , NH_4^+ , total nitrogen [TN], total dissolved nitrogen [TDN], dissolved organic nitrogen, particulate organic nitrogen [PON]) and total phosphorus (TP) removal from synthetic stormwater. Statistically significant removal was only found for NO_3^- and TP. Plant-independent NO_3^- removal occurred 9 months after planting, and then changed to removal only by the least-densely planted *Juncus* treatment. Removal in higher-density *Juncus* plantings was suspected to be limited by preferential pathways created by high root density. *Juncus*' low-density NO_3^- removal success correlates with its high growth rate, root mass and length, and large biomass, matching previous literature. TP removal was plant-independent. Shoot harvesting of one plant of each species after 1 year would remove 0.61 g N. Of the plant species in this study, *Juncus effusus* is most highly recommended for bioretention for its nutrient removal dynamics and year-round green aesthetics. © 2019 Water Environment Federation

• Practitioner points

- Only the one-*Juncus* density treatment had significant NO_3^- removal.
- All *Juncus* treatments as well as non-*Juncus* treatments prevented the PON, TN, or TDN export seen in the No-plants control.
- TP removal was plant-independent.
- *Juncus* had the greatest biomass increase and biomass N. Shoots contain the majority of biomass N for each plant species.
- *Juncus* and *Iris* had high survivorship. Joe Pye had low survivorship. These, and all other study results, need field-scale verification.

• Key words

biological treatment; nitrogen removal; remediation; stormwater; surface water

INTRODUCTION

NONPOINT pollutant sources, such as stormwater runoff, are leading sources of impairment in natural bodies of water around the world, including in the United States (National Water Quality Inventory: Report to Congress, 2004 Reporting Cycle, 2009). Low impact development (LID) techniques treat stormwater runoff near its source. Bioretention is a LID practice that consists of a depression filled with engineered media, covered with mulch, and planted with vegetation. Vegetation plays several roles in bioretention, including aesthetics, community acceptance, maintenance of media infiltration, and (in some cases) pollutant treatment (Muerdter, Wong, & LeFevre, 2018).

Nitrogen (N) and phosphorus (P) are two stormwater pollutants of concern for their contributions to eutrophication in natural waters (Blecken et al., 2010; Conley et al., 2009; Ryther et al., 1971). In bioretention, some plant species provide higher removal of N species than others (Barrett, Limouzin, & Lawler, 2013; Henderson, Greenway, & Phillips, 2007; Milandri et al., 2012; Nocco, Rouse, & Balster, 2016; Read,

Wevill, Fletcher, & Deletic, 2008; Ryciewicz-Borecki, McLean, & Dupont, 2017; Zhang, Rengel, Liaghati, Antoniette, & Meney, 2011). In general, media adsorption is a major P removal mechanism (Bratieres, Fletcher, Deletic, & Zinger, 2008), but some plants have been shown to also significantly contribute to P removal (Barrett et al., 2013; Bratieres et al., 2008; Fowdar et al., 2017; Henderson et al., 2007; Lucas & Greenway, 2008; Milandri et al., 2012; Read et al., 2008; Ryciewicz-Borecki et al., 2017). Work correlating plant traits with bioretention pollution removal performance, for example (Payne et al., 2018; Read, Fletcher, Wevill, & Deletic, 2010), has found that extensive and long roots, high plant biomass, and high growth rate generally correlate with higher N and P removal.

Lacking in the bioretention vegetation literature are many studies comparing single-species plantings with mixed-species plantings. This type of comparison in a single bioretention study has only been conducted in one instance known to the authors, for only 18 weeks (Ellerton, Hatt, & Fletcher, 2012). A mixed planting (*Carex appressa* and *Lomandra longifolia*) was found to create better TN removal than *L. longifolia* alone. Constructed wetland literature provides similar results: High plant diversity and functional group type (legumes, forbs, C3 and C4 grasses) have been correlated with higher ammonium removal rates in constructed wetland microcosms (Ge et al., 2015), and high plant diversity correlates to high N mineralization (Zak, Holmes, White, Peacock, & Tilman, 2003). These results lend further support to the need for the investigation of multispecies effects in bioretention, for longer time periods.

Similarly, the impact of plant density on nutrient removal in bioretention has been understudied. Comparisons of the impact of root structure for different plant species on bioretention nutrient removal (Le Coustumer, Fletcher, Deletic, Barraud, & Poelsma, 2012; Read et al., 2008) suggest that denser vegetation with more root biomass may provide more removal, but the nutrient removal impact of planting density variation of a single species in bioretention is essentially unexplored (Muerdter et al., 2018). Interestingly, in a constructed wetland study, lower plant density (50% surface cover) was associated with greater microbial diversity and greater nitrate removal (Ibekwe, Lyon, Leddy, & Jacobson-Meyers, 2007).

Finally, the potential nutrient impact of shoot harvesting in bioretention requires more data. Shoot harvesting permanently removes N from the bioretention system, which could otherwise be returned to the system through leaf drop (Muerdter et al., 2018).

Given these research needs, this study sought to:

1. Compare three under-evaluated bioretention plant species, in bioretention container monoculture, to determine which species produce the most N (NO_3^- , NO_2^- , NH_4^+ , total nitrogen [TN], total dissolved nitrogen [TDN], dissolved organic nitrogen [DON], particulate organic nitrogen [PON]) and total P removal.
2. Test the impact of varied *Juncus effusus* plant density on N and P removal. Because increased root mass generally correlates with higher nutrient removal in bioretention

literature, the densest planting was expected to produce the most removal.

3. Compare monoculture plantings and multispecies plantings for their impact on N and P removal. Removal was expected to be greatest in the multispecies planting due to varying phenology and uptake of the different plant species.
4. Quantify plant biomass and biomass percent N change over a year to determine how much N would be removed with shoot harvesting.

While limited to one year of container data, the results of this study inform selection of bioretention plant selection for N and P removal and provide an increased understanding of both multispecies plantings and plant density effects.

METHODOLOGY

The study was conducted in the University of Maryland Research Greenhouse, with similar conditions to outdoor, ambient lighting. A temperature regime (S.I.A.) was derived from College Park, Maryland, temperature data.

Containers

Container details are given in SI.I.B. Briefly, 13.2-L containers were assembled beginning with ~30 cm² of a single layer of gravel directly above the drainage hole. 2.5 cm of washed sand was placed on top of the gravel, followed by 18.5 cm of bioretention soil media and 1.3 cm of mulch. 5.0 cm of headspace was left for water ponding. Container size was limited by the amount of media, mulch, and gravel available.

Before vegetation planting, freshly assembled containers were found to produce very turbid effluent when DI water was run through them. To produce effluent clear enough for the analysis of N species, assembled buckets were flushed with 224 cm (140 L) of DI water followed by 26.9 cm (16.8 L) of synthetic stormwater.

Plants used

Plant species were selected after consultation with local bioretention professionals (Mary Travaglini, personal communication, 2013), design manuals (Department of Environmental Resources of Prince George's County Maryland, 2007), and the bioretention literature. *Iris versicolor* (monocot, common name: Iris), *Juncus effusus* (monocot, common name: common rush), and *Eutrochium dubium* (dicot, common name: Joe Pye) were chosen for their past survivorship, growth, and aesthetics in Maryland bioretention (Center for Watershed Protection, & Maryland Department of the Environment, 2009; Muerdter, Özkök, Li, & Davis, 2016; Montgomery County, Maryland Department of Environmental Protection, 2016). Additionally, they were chosen for their variety of root characteristics, biomass, and growth rate, which (see Introduction) are all expected to influence N and P uptake. They were also selected by their variety of bloom and growth phenology, which was expected to provide more continuous aesthetic interest and possibly influence nutrient uptake timing during the year. These three plants

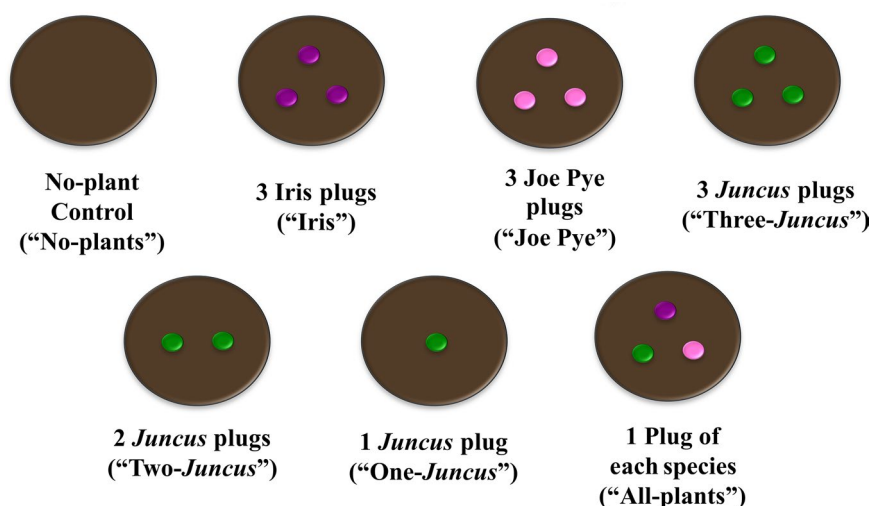


Figure 1. Treatments used in the study, as created by different planting schemes. Throughout the study, any unplanted plant species growing in the containers were removed as soon as they were seen.

have infrequent (*Juncus effusus*) or nonexistent (*Eutrochium dubium* and *Iris versicolor*) occurrence in the bioretention literature. At the time of this study, *Juncus effusus* had only been evaluated in the literature in a bioretention cell co-planted with seven other species (Hunt, Smith, Jadlocki, Hathaway, & Eubanks, 2008). Since this study, work has been published on *Juncus effusus* co-planted with 15 other species in bioretention cells exposed to high nutrient runoff in North Carolina (Turk, Kraus, Hunt, Carmen, & Bilderback, 2017) and on a mono-species *Juncus effusus* column study with stormwater-relevant concentrations in China (Wang et al., 2017). Other *Juncus* species have been evaluated in bioretention: *Juncus amabilis* and *Juncus flavidus* (Read et al., 2010, 2008) were among the most effective tested plants at N and P removal per unit root mass.

After flushing with DI water and synthetic stormwater, containers (except the no-plants treatment) were planted with vegetation (plugs obtained from Signature Horticultural Services, Inc., Freeland, MD) in study week one. Before planting, as much nursery soil as possible was gently removed from the roots by hand. The plug was rinsed with DI water twice and gently massaged by hand to remove as much of the soil as possible while leaving the roots intact, and then planted. A no-vegetation control along with planted treatments, as shown in Figure 1, was assembled. Each of the seven treatments was replicated four times, for a total of 28 containers arranged in four rows of seven.

Media samples

Media samples were taken at the beginning and end of the study and analyzed for percent N through combustion at the University of Delaware Soil Testing Program. At the beginning of the experiment, composite media samples before washing with DI and stormwater ($n = 2$) and after washing ($n = 2$) were taken. The after-washing samples were collected from planting depth (0 to ~6.4 cm below mulch) from 11 and 17 containers, respectively, for each sample, during planting. At the end of the experiment, media samples were collected from the same depth

range from three containers each of: one-*Juncus*, three-*Juncus*, No-plants, and Iris.

Applied water

The plantings were established with 1.6 cm of tap water applied per day for 14 days, per the Maryland Stormwater Design Manual (Center for Watershed Protection, & Maryland Department of the Environment, 2009). Subsequently, except for two tap waterings in weeks 3 and 4, and a second establishment period after Joe Pye replanting in week 16, only synthetic stormwater was applied. The synthetic stormwater consisted of 0.56, 0.5, and 0.3 mg N/L NaNO_3 , glycine, and NH_4Cl , respectively; 0.2 mg P/L NaH_2PO_4 , and 0.01 M NaCl. Stormwater components and concentrations were informed by measurements of stormwater from two nearby locations (Li & Davis, 2009). Synthetic stormwater was created in a 167-L container using the chemicals listed above and DI water, mixed with a laboratory stirrer.

In week 16 through the end of the study, the following chemicals were also added in order to apply, per three months of stormwater applications, the mass of each chemical required to make up the deficit between tested and optimal concentrations for plant health: 6.7 mg K/L KCl, 0.78 mg Mg/L from MgCl_2 and MgSO_4 , 0.68 mg S/L from MgSO_4 and ZnSO_4 , 0.045 mg B/L $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \times 8\text{H}_2\text{O}$, and 0.028 mg Zn/L ZnSO_4 . This decision was made to ensure plant survival in the experiment by providing chemicals that are typically present in stormwater (Hasenmueller & Criss, 2013; McKenzie, Money, Green, & Young, 2009) but had not been applied up to that point. No storm events were analyzed for N or P (S.I.C.) between the times when Joe Pye chlorosis was first noticed and when the Joe Pye plants were replaced with new plants and the above chemicals added to the stormwater. Thus, poor Joe Pye health is not expected to influence the N and P results, except for the Joe Pye plants being slightly younger than the other two plant species after the replanting.

Stormwater was applied to mimic natural Maryland precipitation (Kreeb & McCuen, 2003), generally twice a week

(watering schedule in SI.I.D.). The total applied stormwater runoff depth delivered during this study was 1,674 cm, corresponding to a total simulated stormwater rainfall depth of 93.0 cm.

Effluent rates

Effluent flow rate measurements (quantified as the elapsed time between the first application of stormwater and the time when 2.54 cm of effluent was collected) were made ~monthly; typically one container of each treatment was measured. Beginning about one month into the study, slow drainage of some containers was noted. In week 29 of the experiment, 15 of the containers, representing all treatments, were draining slowly enough so that not all stormwater could be immediately applied. Biofilm growth at the drainage hole was suspected. Therefore, in week 29, an acid bath-cleaned 1,000- μ L pipette tip was inserted into the hole at the bottom of each container as far as possible (about 4.5 cm) and then removed, three times. A new pipette tip was used for each container. This procedure restored drainage rates so that all water could be immediately applied. The pipette procedure was repeated ~every 2 weeks for the remainder of the experiment, maintaining drainage rates without further problems.

Water samples

Collection. Seven simulated storms were sampled and analyzed for at least one N species and/or TP (SI.I.E.), using methods given in SI.I.F. Those storms were selected for analysis due to their distribution throughout the study and their varying storm depths. Influent samples were collected for each batch of stormwater. Effluent samples were collected either as composite or as discrete samples. Composite samples were collected once all effluent has drained through the containers. The collected effluent was stirred with a clean rubber spatula and a 125-ml sample bottle was submerged into the effluent with a gloved hand until the bottle was full, minus headspace for freezing. For a discrete sample, a sample bottle was placed directly below the effluent funnel while stormwater was applied to the top of the container. Thus, composite samples were meant to capture the entire storm event, whereas discrete samples measured the instantaneous impact of the container on the effluent—a “first-flush” measurement if taken at the beginning of a storm.

Statistics

Unless otherwise noted, either (a) an approximative k-sample permutation test was performed with the coin package in R, using the discrete method to find the p -value (significance declared at $p \leq 0.05$) for each treatment/control, or (b) Kruskal–Wallis followed by Dunn’s test was performed with the dunn.test package in R, with significance declared at $p \leq 0.05$.

Plant biomass collection and measurement

Plant biomass dry weight and percent N were measured at the study’s beginning and end (SI.II.A and B). Dry mass and percent N were used to find mass of N in plant biomass using the SI.V.A. equation. Percent P in plant biomass was not measured.

RESULTS AND DISCUSSION

Plant survivorship

All of the *Juncus* except for one plug survived, and all of the Iris survived. In contrast, all of the Joe Pye plants displayed severe chlorosis early in the study (plant health timeline is given in SI.I.C.). This is likely due to the lack of nutrients, such as zinc, which would be present in natural stormwater (McKenzie et al., 2009) and were added to the stormwater after the Joe Pyes were replanted in week 16. These replacement plants survived well until they died back for the winter as expected. Only 56% of these replanted Joe Pyes sprouted again in the spring, however, though they did grow without chlorosis until the end of the study. This lack of resprouting was an unexpected result, given the high percent coverage of Joe Pye in a local bioretention cell (Muerdter et al., 2016). Joe Pye demonstrated initial survivorship difficulty even in that field cell, however: 20% were replaced after failing in the first year after planting (Craig Carson, personal communication, 2013). In container, size may also have contributed to plant failure via more media drying than would occur in a much larger field cell.

Hydrology

As noted in the methods, beginning about 1 month into the study some of the containers began to drain more slowly. Increased hydraulic retention time is generally associated with higher TN removal (Lucas & Greenway, 2011). In this study, slow drainage and corresponding high hydraulic retention time persisted for at least some of the containers (15 of 28 containers at a maximum) from week 4 through week 29: August 2014 through the end of January 2015 (timeline of slow drainage, storm events, etc., provided in SI.I.C.). This time period includes two of the storms in which a significant difference occurred between the influent and the effluent from at least one treatment (Figure 2b,e): 1/14/15 for TP and 10/31/14 for TN-N. The same TP trend is seen in 4/2/15 results, after slow hydraulic retention times had been fixed, as in 1/14/15 results, so hydraulic retention time does not appear to influence this result. In the 10/31/14 TN results, no treatment had significant TN removal, as would be expected with higher retention times in some containers. Thus, the two dates of significant results during the time period of higher retention times do not appear to be influenced by hydraulics. The other dates with significant data (Figure 2) were outside of the time period affected by slow drainage.

Objective 1: P and N influent and effluent concentrations

Significant N or P removal. Significant removal by a treatment vis-à-vis the influent was only found for TP and NO_3^- . TP removal in this study appears to be plant-independent (Figure 2b) and was first observed in the study’s sixth month (1/14/15, sampled discretely at 0.35 cm applied), as well as on 4/2/15 (a 1 cm composite sample). The removal rates for the two storms are very similar: 58%–79% (depending upon treatment) for 1/14/15 and 57%–75% for 4/2/15. Thus, P removal in these two

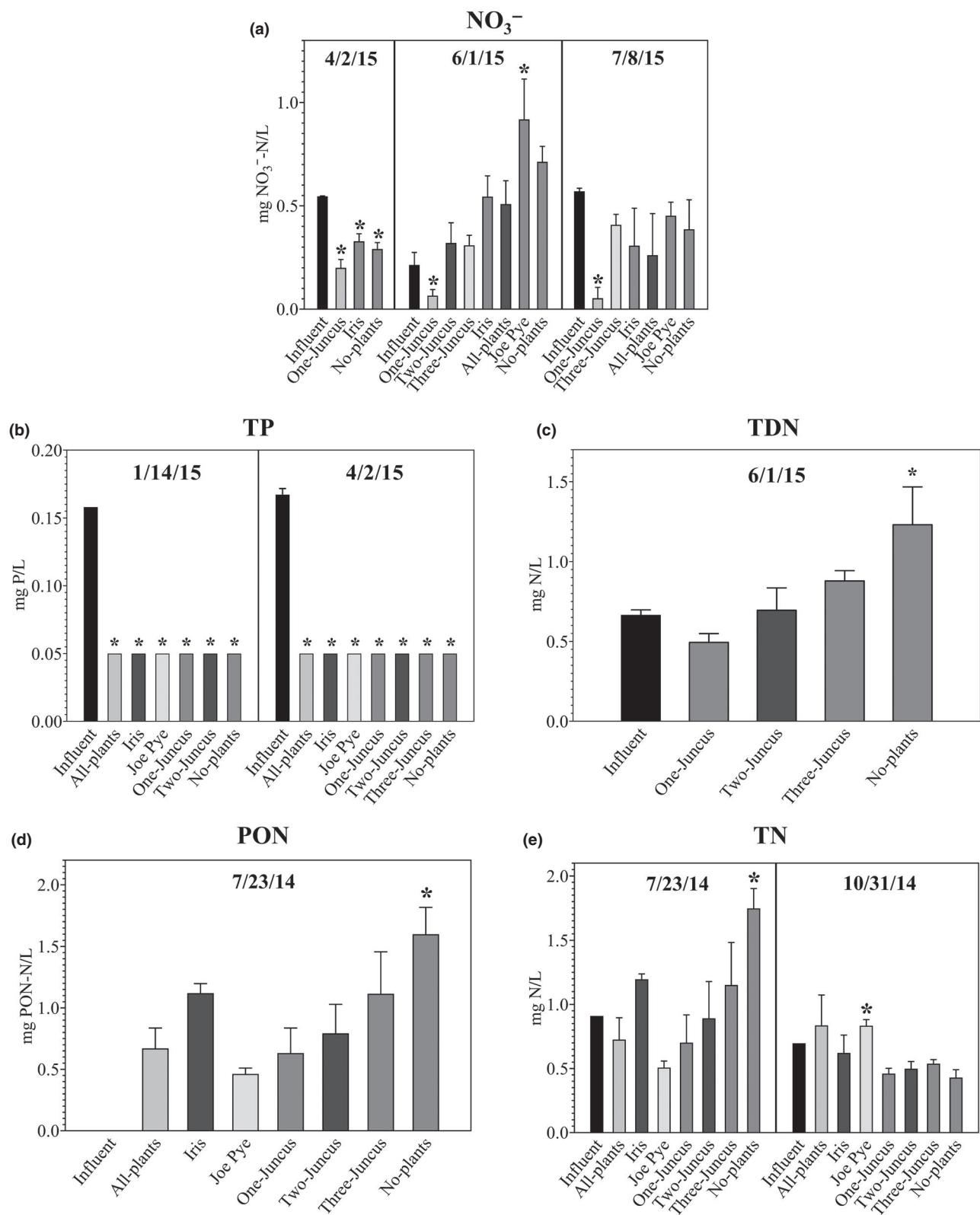


Figure 2. Sampled storms in which a significant ($p \leq 0.05$) difference in a measured N or P species was found between the influent and at least one treatment, indicated by an *. $n = 1-3$ for influent; $n = 2-4$ for effluent, usually $n = 4$. Error bars are $\pm$ one standard error. Storm event details are given in the Supporting Information. All treatments and control concentrations for (b) are one half of the detection limit.

Table 1. Plant biomass and N in plant biomass results

PLANT SPECIES	BIOMASS DATA (DRY WEIGHTS)				N DATA			
	TREATMENT(S) SAMPLED	AVERAGE CHANGE IN ROOT BIOMASS (G)	AVERAGE CHANGE IN SHOOT BIO-MASS (G)	AVERAGE INCREASE IN TOTAL BIOMASS (±ONE STANDARD DEVIATION)	MASS N ACCUMULATED IN ONE PLANT (ROOTS AND SHOOTS) (±ONE STANDARD DEVIATION) (G)	MASS N ACCUMULATED IN ONE PLANT (G)/MASS OF PLANT BIOMASS (G)	FRACTION OF APPLIED STORMWATER N ACCUMULATED IN SHOOTS	FRACTION OF APPLIED STORMWATER N ACCUMULATED IN PLANT (ROOTS AND SHOOTS)
Joe Pye	Joe Pye	+0.090 (in 38 weeks)	+2.4 (in 38 weeks)	+1.3 (±1.6)	0.064 (±0.052)	0.01	0.060	0.067
<i>Juncus</i>	One- <i>Juncus</i> and all-plants	+19 (in 54 weeks)	+37 (in 54 weeks)	+32 (±14)	0.50 (±0.28)	0.01	0.42	0.53
Iris	Iris and all-plants	+3.9 (in 54 weeks)	+3.0 (in 54 weeks)	+3.3 (±2.0)	0.046 (±0.033)	0.01	0.031	0.048
SUM of one plant of each species	See above	+23	+42	+20	0.61	0.03	0.51	0.65

Note. Plant biomass and N mass results from the time of planting ($n = 3$ for all plant species) and end of the study ($n = 3$ for Joe Pye and $n = 6$ for *Juncus* and Iris). Due to the failure of initial plantings, Joe Pye data are for 38 weeks rather than the full 54 weeks of the study. Calculation details are given in Supporting Information V.A and V.B.

storms seems to proceed at a similar rate throughout the storm. The P removal taking time to develop between the beginning of the study and the 1/14/15 storm may be due to export of P from media organic matter breakdown (LeFevre et al., 2015) initially counteracting any P removal. These plant-independent results agree with other studies, suggesting that bioretention dissolved P removal is controlled by media chemistry (Hunt, Davis, & Traver, 2012; LeFevre et al., 2015; Li & Davis, 2016) but contrast with bioretention studies that demonstrate P removal significantly impacted by plant selection (Barrett et al., 2013; Fowdar et al., 2017; Henderson et al., 2007; Lucas & Greenway, 2008). Significant plant selection impacts may be found if plant species besides those used in this study were examined.

NO_3^- was significantly removed for three events (Figure 2a), including two instances where removal was not seen in the unplanted control. Initially, in early spring (4/2/15), NO_3^- removal was plant-independent, likely driven by the microbial community (Hunt et al., 2012; LeFevre et al., 2015). Significant NO_3^- removal occurred in a plant-dependent manner on 6/1/15 (77% removal, composite 0.7 cm sample) and 7/8/15 (91% removal, first-flush sample), only in the one-*Juncus* treatment. The slightly higher removal in the 7/8/15 storm may suggest that removal declines with added stormwater, as application rate exceeds plant NO_3^- uptake rate. The *Juncus*-mediated removal in these two events may be a seasonal trend, and/or it may result from the plants reaching a certain biomass level several months after planting. *Juncus* had significantly longer roots than Joe Pye in all measured treatments (all-plants, Joe Pye, and one-*Juncus*, $p = 0.001$ – 0.03), a trait correlated to better N removal in previous bioretention work (Payne et al., 2018; Read et al., 2010). Greater root mass has been shown to correlate with greater denitrifying microbial abundance in constructed wetlands (Ge et al., 2011), which may correlate to this study in which *Juncus* had the greatest increase in root mass of the three species during the study (Table 1) and was the only plant species to create NO_3^- removal.

NO_3^- or NO_x removal with one plant species vis-à-vis others was previously documented in planted bioretention containers (Barrett et al., 2013; Bratieres et al., 2008; Hunt, Lord, Loh, & Sia, 2015; Milandri et al., 2012; Read et al., 2008; Zhang et al., 2011), but has not been previously demonstrated to be due to plant density. This result in this study is believed to be due to plant density effects interacting with container size to create preferential pathways (see Objective 2 results below).

Significant N or P export. The significant ($p = 0.01$) export of TN in the Joe Pye treatment on 10/31/14 (Figure 2e) was likely due to media N dislodged during Joe Pye replanting two days earlier, rather than a true plant-mediated effect. Export is defined as a greater concentration of a measured compound in the effluent than in the influent.

Planted treatments of all types did occasionally prevent the significant export (Figure 2c–e) of PON, TN, or TDN found in the no-plants control. Additionally, NO_3^- export prevention was also found in multiple planted treatments on 6/1/15 (Figure 2a). While only the one-*Juncus* treatment created significant removal of NO_3^- , the

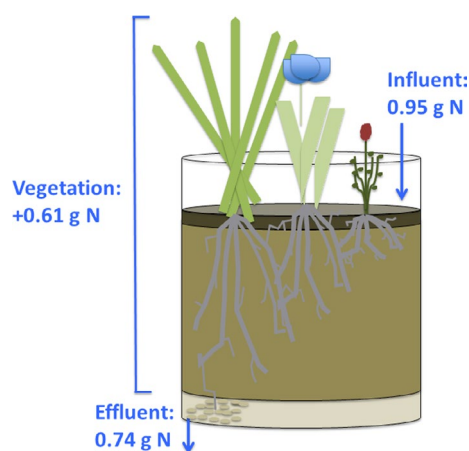


Figure 3. For an all-plants treatment: Average mass of N applied in the 1,674 cm³/1,045 L simulated stormwater applied per mesocosm during the study (calculated in SI.V.B., note that tap water N is not included but is estimated in SI.I.D. to only be 3% of the total applied N and therefore to be insignificant), accumulated in the plants (calculated in SI.VI.), and exported in effluent (calculated in SI.V.C.), during the course of the study, as g N. The vegetation value is a sum of an average Iris and average *Juncus*, both from the all-plants treatment, as well as a Joe Pye plant from the Joe Pye treatment (with only 38 weeks of growth for Joe Pye vis-à-vis 54 weeks for the other two plant species). Media percent N varied little between the beginning and end of the study (SI.IV.); therefore, media N was not included in this figure.

higher-density *Juncus* treatments, along with the Iris and all-plants treatments and no-plants control, prevented the export of NO_3^- seen in the Joe Pye treatment (Figure 2a). By this time, the Joe Pye treatment had one plant instead of the original three, expected to limit uptake. Previous bioretention studies (Davis, Shokouhian, Sharma, & Minami, 2001, 2006; Dietz & Clausen, 2005; Hsieh, Davis, & Needelman, 2007; Hunt, Jarrett, Smith, & Sharkey, 2006; Line & Hunt, 2009) have also found NO_3^- export, typically attributed to between-storm microbial nitrification.

Overall TN removal. Media percent N was similar between the beginning and end of the study, and thus appears unimportant in N dynamics (SI.IV.). The overall average removal of 22% TN between influent and effluent agrees fairly well with previous field bioretention work (Li & Davis, 2014), which showed a 9% net TN reduction. It agrees very well with the previous *Juncus effusus* study in China, with a ~20% TN removal rate with stormwater and media conditions that most closely resemble that of this study (Wang et al., 2017).

Objective 2: plant density effects

Significant plant-related removal of N or P was only found in the lowest-density *Juncus* planting, for NO_3^- . A hypothesis for this plant density effect is that the additional root structure of multiple plants creates higher effluent flow rates and thus less stormwater contact time with the container components, which is especially likely given the depth of media: 18.5 cm versus 30–70 cm in other studies such as Barrett et al. (2013) and Bratieres et al. (2008). This hypothesis is supported by flow data. The 7/13/15 (after the slow drainage problem had

been resolved) average effluent flow rates for the two-*Juncus* treatment, three-*Juncus* treatment, and no-plants control were 160%–180% of the one-*Juncus* rate. The slow effluent rate in the one-*Juncus* treatment corresponded with more NO_3^- removal. The faster flow rate in the higher-density treatments would potentially negate any increase in N uptake from increased root surface area from additional plants. Further work, including information on *Juncus effusus* performance in deeper media, is needed.

Objective 3: effect of plant diversity

Contrary to the expectation that multiple species blooming and growing at different times would create greater overall uptake, the all-plants treatment had no significant removal events. This may be due to the individual plant species used. Other multispecies combinations may produce better nutrient removal results.

Objective 4: plant biomass and N content increase, shoot harvesting recommendations

The sum of above- and belowground biomass increase in an average *Juncus* plant during the study was more than an order of magnitude greater than the increase for Iris or Joe Pye (Table 1, full data in Supporting Information). The Joe Pye value represents only 38 weeks of growth vis-à-vis 54 weeks for *Juncus* and Iris, but even if given the additional 16 weeks, the Joe Pye plant would likely not produce as much biomass as *Juncus* or even as Iris.

On average, one *Juncus* plant accumulated 0.50 ± 0.28 g N in its biomass during the study, vis-à-vis 0.064 ± 0.052 g N for one Joe Pye and 0.046 ± 0.033 g N for one Iris. The variation in percent N between species at the end of the study (2.4-fold for roots and 2.2-fold for shoots) was less than the variation in plant biomass increase between species (>8-fold). Thus, plant biomass is the primary factor determining the total N in plant biomass.

Thus, over the course of this study 0.61 g of N accumulated in the plant biomass (of one average plant of each species), a large number given that the influent N applied during the study was 0.95 g (SI.V.B., Figure 3, or as mass loading/area in SI.III.C.) and unexpectedly high given the lack of significant TN removal in any individual storm event. Besides direct N uptake from stormwater, other mechanisms for N sequestration into plant biomass must have occurred, for example, microbial N fixation (Nelson, Martiny, & Martiny, 2016). Nitrogen fixation has been reported to occur in the *Juncus effusus* rhizosphere (Jagnow, 1983; USDA Plant Guide Soft Rush *Juncus effusus*, n.d.). In a wetland field setting, the fairly closely related (Drabkova, Kirschner, Seberg, Petersen, & Vlcek, 2003) *Juncus balticus* was found to fix N at 15°C at a rate that can be extrapolated to ~1.8 g N/year (Tjepkema & Evans, 1976). Thus, a year-long accumulation likely through both nitrogen fixation and uptake from stormwater of 0.50 g N by *Juncus effusus* (SI.VI.) is not unreasonable. Despite fixation potentially introducing an additional source of nitrogen, *Juncus* was the only plant to create significant nitrogen (NO_3^-) removal from the simulated stormwater in this study. This is in agreement with previous

work, suggesting that *Juncus effusus* rhizosphere nitrogen fixation can be depressed but not eliminated by nitrogen addition (Ogan, 1982). Additionally, nitrogen-fixing plants correlated with high N removal in constructed wetland mesocosms (Ge et al., 2015).

In this study, cutting all aboveground biomass would remove the majority of biomass N ($\geq 80\%$) for all three plant species (Table 1). Harvesting *Juncus* shoots would permanently remove the most N of any of the three plant species, due to the amount of N in its shoots. If the plants survive and grow back at a similar rate (not determined in this study), shoot harvesting can permanently remove N from the system.

CONCLUSIONS

The results of this study need to be replicated in larger containers and/or at field scale to ensure the practicability of these recommendations at field scale. In this study, the lowest density *Juncus* treatment produced the best N removal. Suspected preferential pathways in Two- and Three-*Juncus* treatments in this study are expected to change in a larger container or field setting, impacting plant density results. This emphasizes the importance of media depth in bioretention nutrient functioning. Additional useful information can be taken from these study results.

First, they provide information on bioretention plant survivorship: Both *Juncus* and *Iris* had high survivorship, and Joe Pye had low survivorship. This matches reported field results of Joe Pyes struggling with establishment the first year after planting.

Second, these results confirm past bioretention and constructed wetland vegetation work on high plant biomass, root length/mass, and growth rate correlating with high nutrient removal (Barrett et al., 2013; Bratieres et al., 2008; Ge et al., 2011; Payne et al., 2018; Read et al., 2010). In this study, *Juncus* was the only plant that created significant N or P removal (i.e., of NO_3^-). *Juncus* had the most biomass (Table 1), greatest growth rate (Table 1), and longest root length (S.I.I.C) of the three species tested. These data further support the importance of these traits in bioretention nutrient performance and provide information to practitioners looking for NO_3^- removal in bioretention.

This study also demonstrates that *Iris versicolor* and Joe Pye (*Eutrochium dubium*) are not successful at nutrient removal in bioretention containers in the first year after planting. Both plant species prevented PON and TN export seen in the No-plants control on one date early in the study, but otherwise did not create significant nutrient results. They do provide aesthetic value, including flowers. While the nutrient results should be verified in the field, they provide useful preliminary information for practitioners when considering bioretention plant selection.

Likely because of the specific plant species selected, the All-plants treatment did not outperform individual plant species treatments, unlike constructed wetland studies (Ge et al., 2015; Zak et al., 2003). Combinations of known individually high-performing species may produce greater nutrient removal in multi-species plantings.

Another potentially useful piece of information for practitioners is that none of the three plant species created significant

TP removal. TP removal was plant-independent and likely media-driven. Thus, other plant species need to be considered to increase media TP removal.

Finally, this study provides information on the amount of N removed by plant shoot harvesting, which is underdocumented in the literature. Plant shoot harvesting for all three species after one year would remove the majority of plant biomass N, with *Juncus* representing the greatest removal amount.

ACKNOWLEDGMENTS

The authors acknowledge the Maryland State Highway Administration for funding support; Stancills, Inc., for materials donations; and Katie Judd, Marya Anderson, and Doris Ihejirika, Sydney Wallace, Meghan Holbert, and Shaun Faulkner for laboratory and greenhouse assistance. The authors also thank Drs. Dong Liang and Richard McCuen for statistical guidance, Danielle Kennedy and Dr. Gregory H. LeFevre for manuscript review, and Dr. Whitney Griffin and Clark DeLong for experimental design advice.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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